Instance Segmentation

Instance Segmentation

Classification

CAT

No spatial extent

Semantic Segmentation

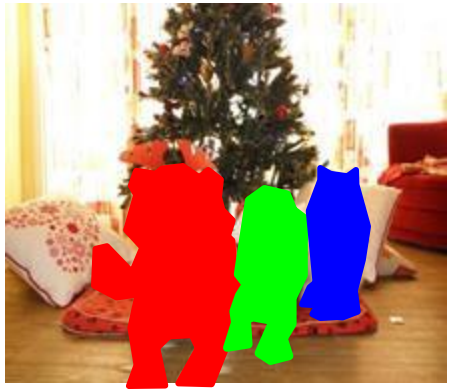
GRASS, CAT, TREE, SKY

No objects, just pixels

Object Detection

DOG, DOG, CAT

Instance Segmentation



DOG, DOG, CAT

Multiple Object

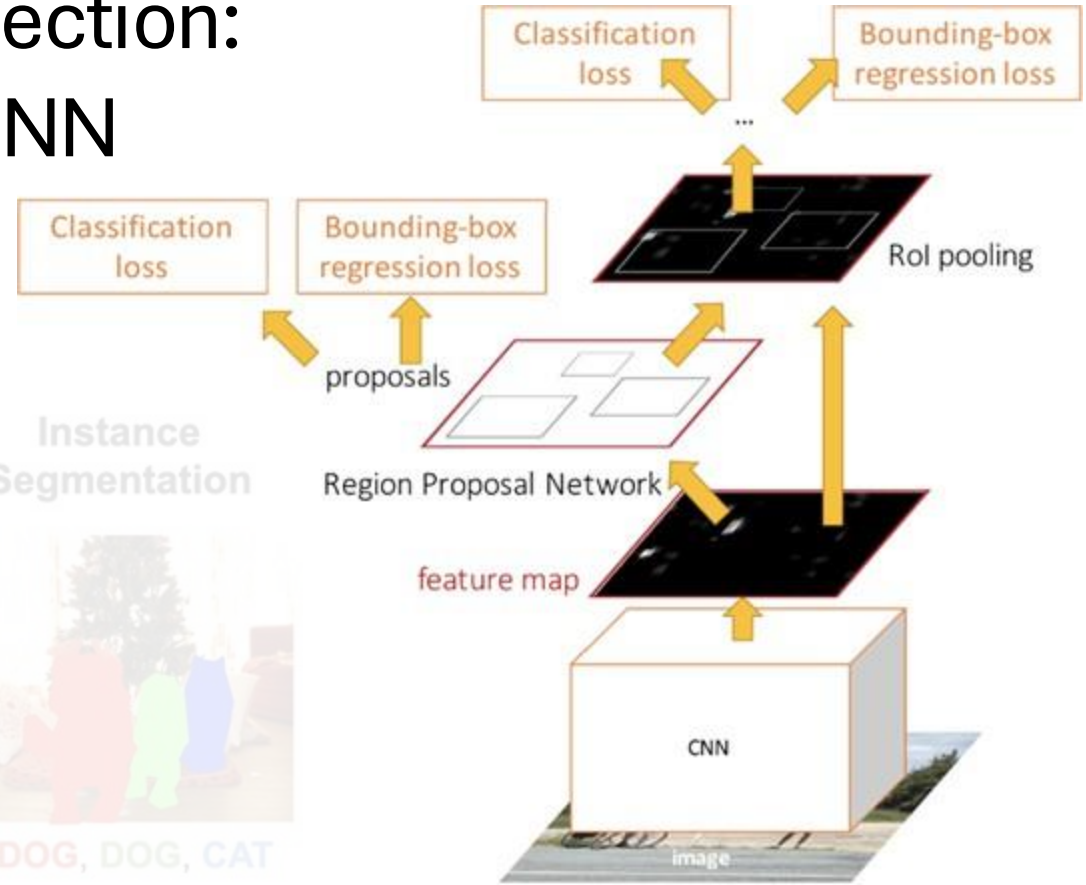
Object Detection: Faster R-CNN

Object Detection

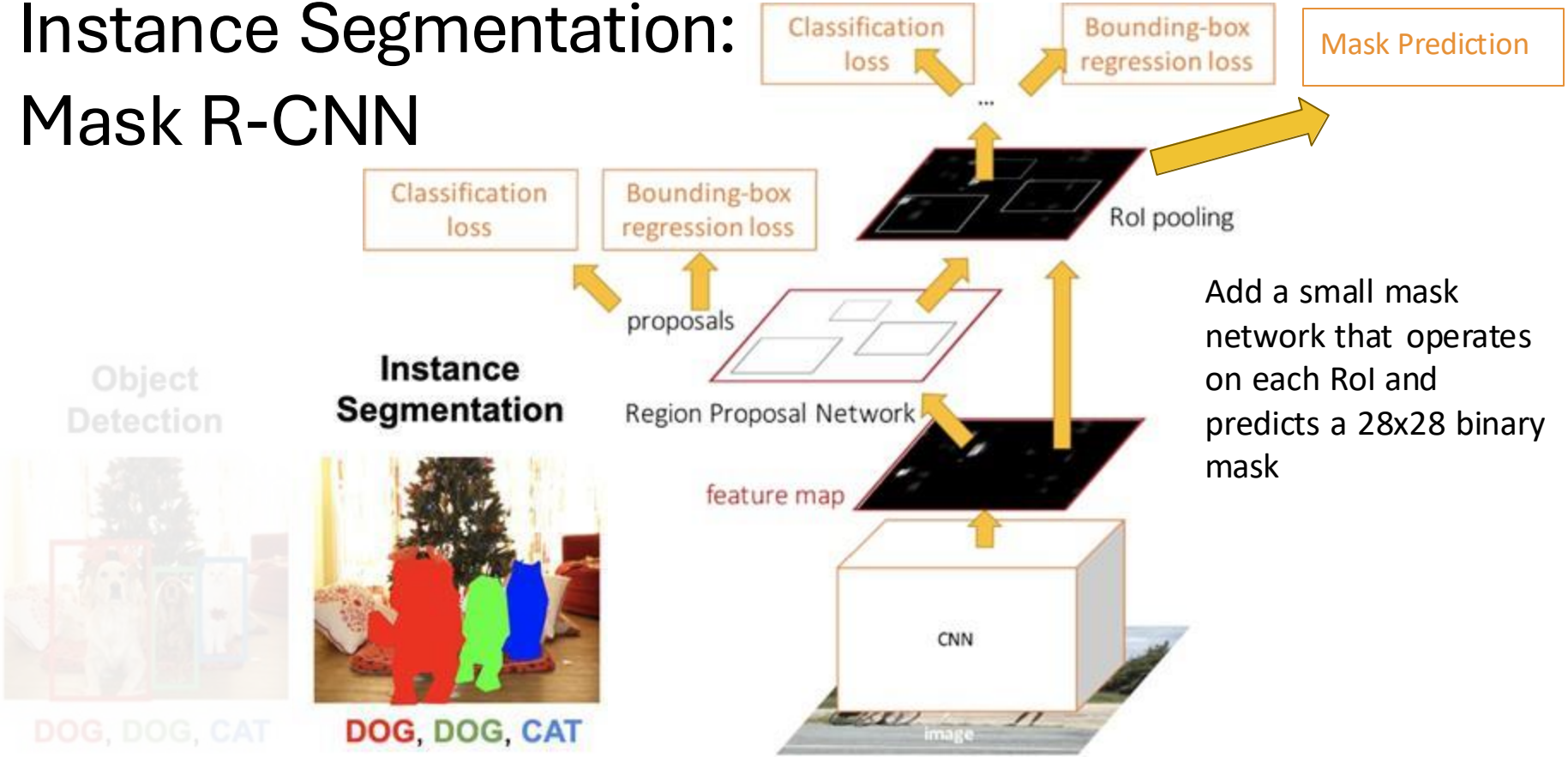
DOG, DOG, CAT

Instance Segmentation

DOG, DOG, CAT



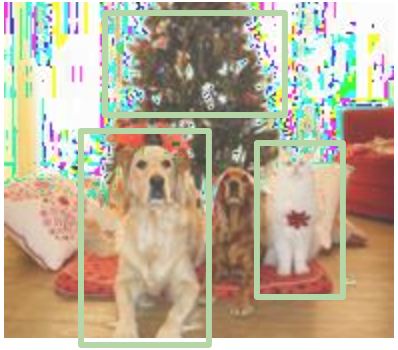
Instance Segmentation: Mask R-CNN



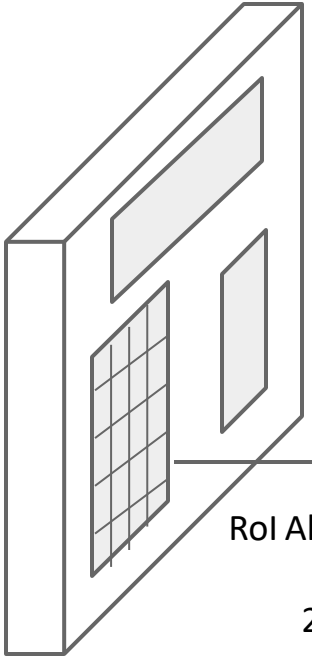
Add a small mask network that operates on each RoI and predicts a 28x28 binary mask

He et al, "Mask R-CNN", ICCV 2017

Mask R-CNN

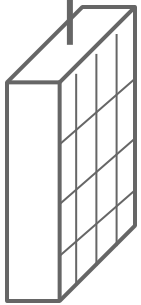


CNN
+RPN



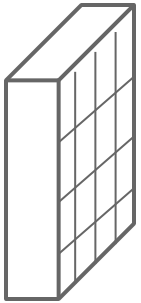
RoI Align

Classification Scores: C
Box coordinates (per class): 4 * C



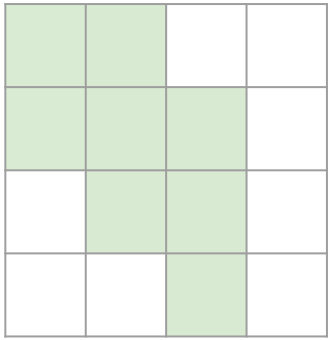
256 x 14 x 14

Conv



256 x 14 x 14

Conv

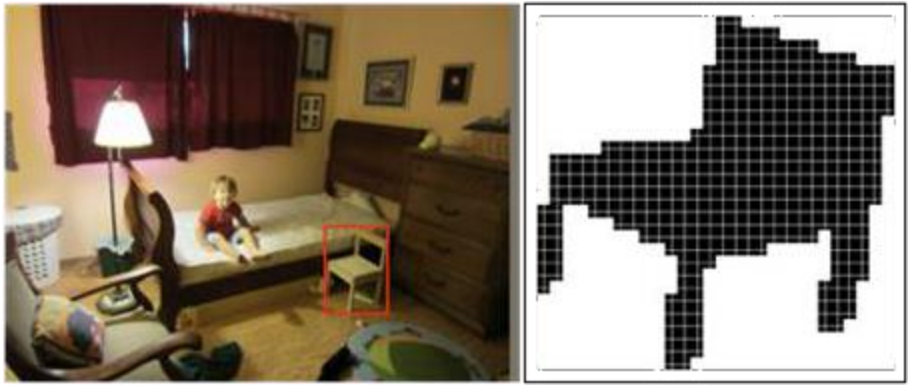


Predict a mask for each of C classes

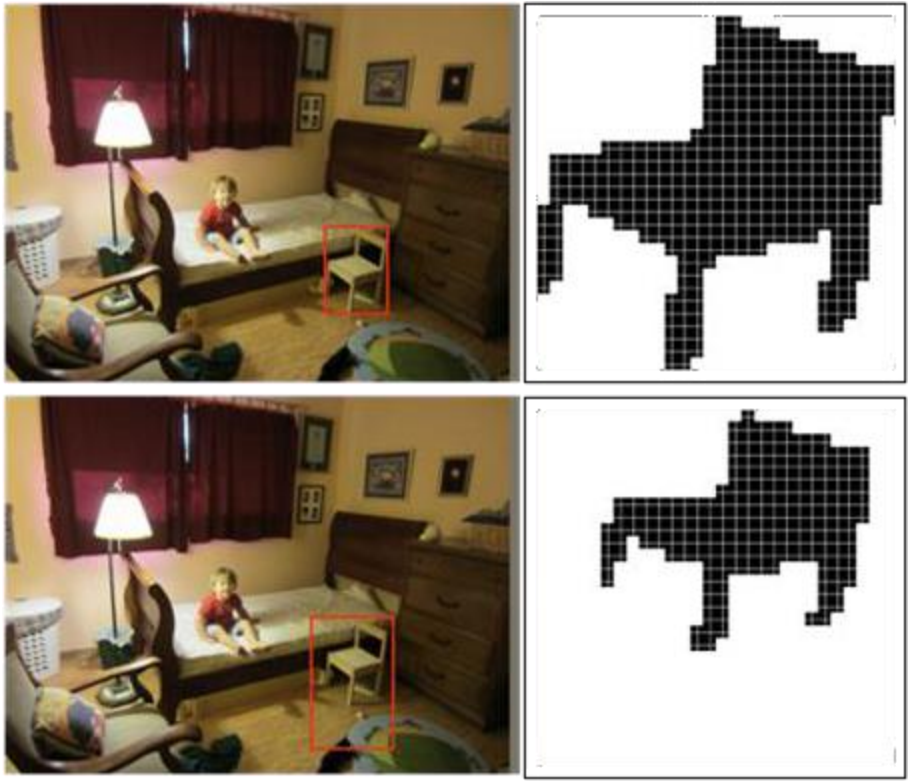
C x 28 x 28

He et al, “Mask R-CNN”, arXiv 2017

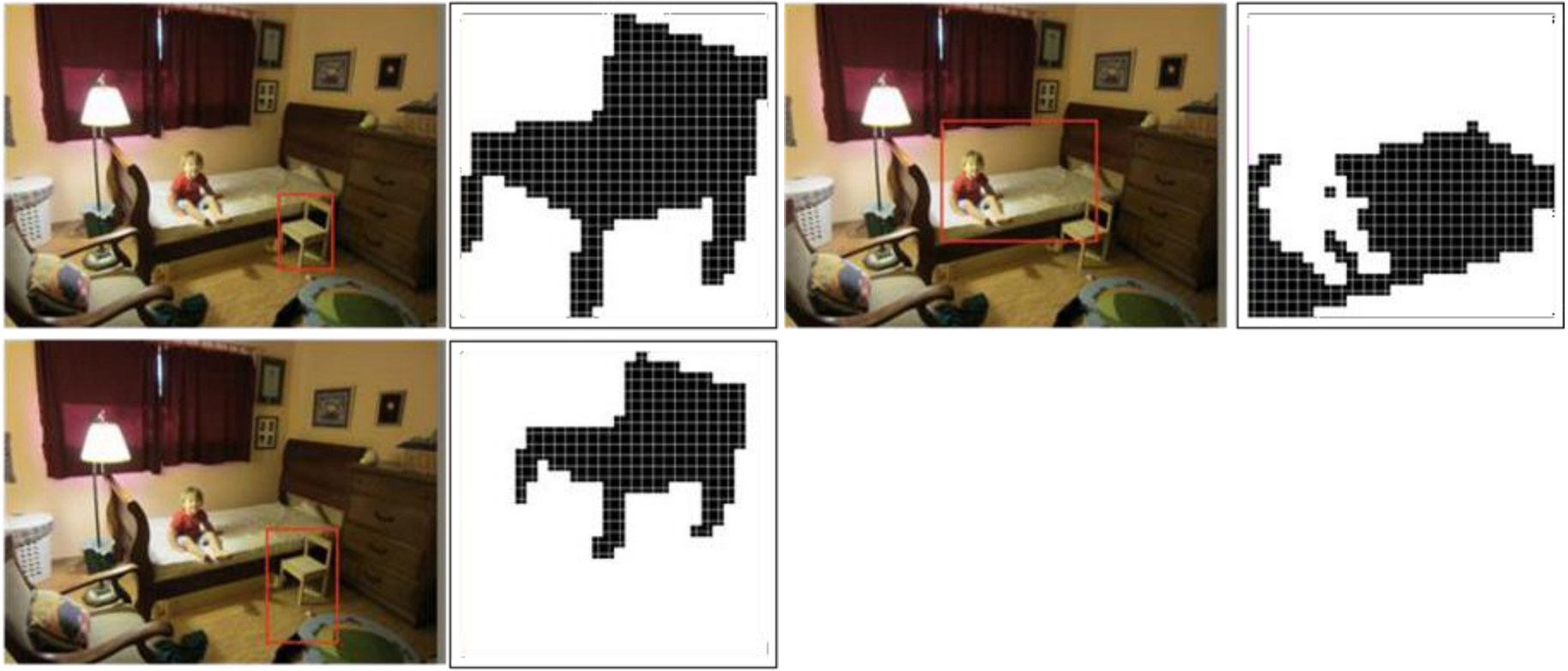
Mask R-CNN: Example Mask Training Targets



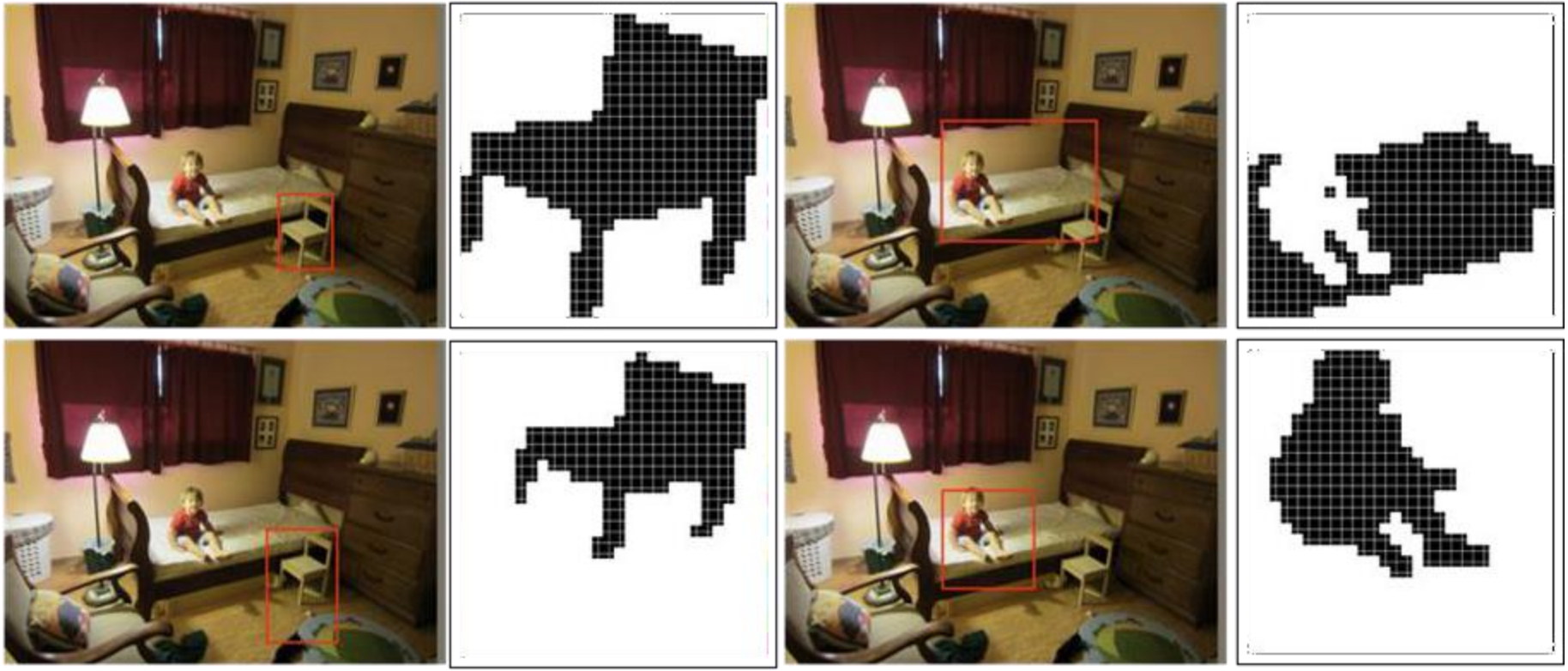
Mask R-CNN: Example Mask Training Targets



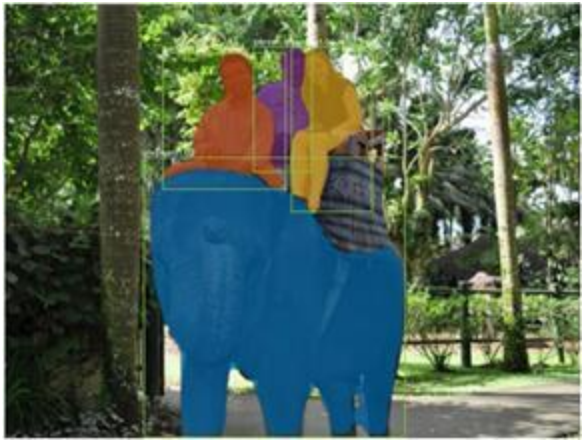
Mask R-CNN: Example Mask Training Targets



Mask R-CNN: Example Mask Training Targets

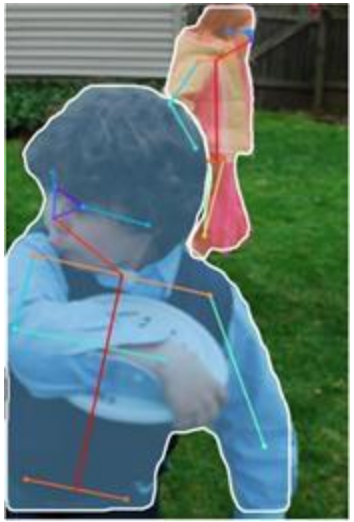


Mask R-CNN: Very Good Results!



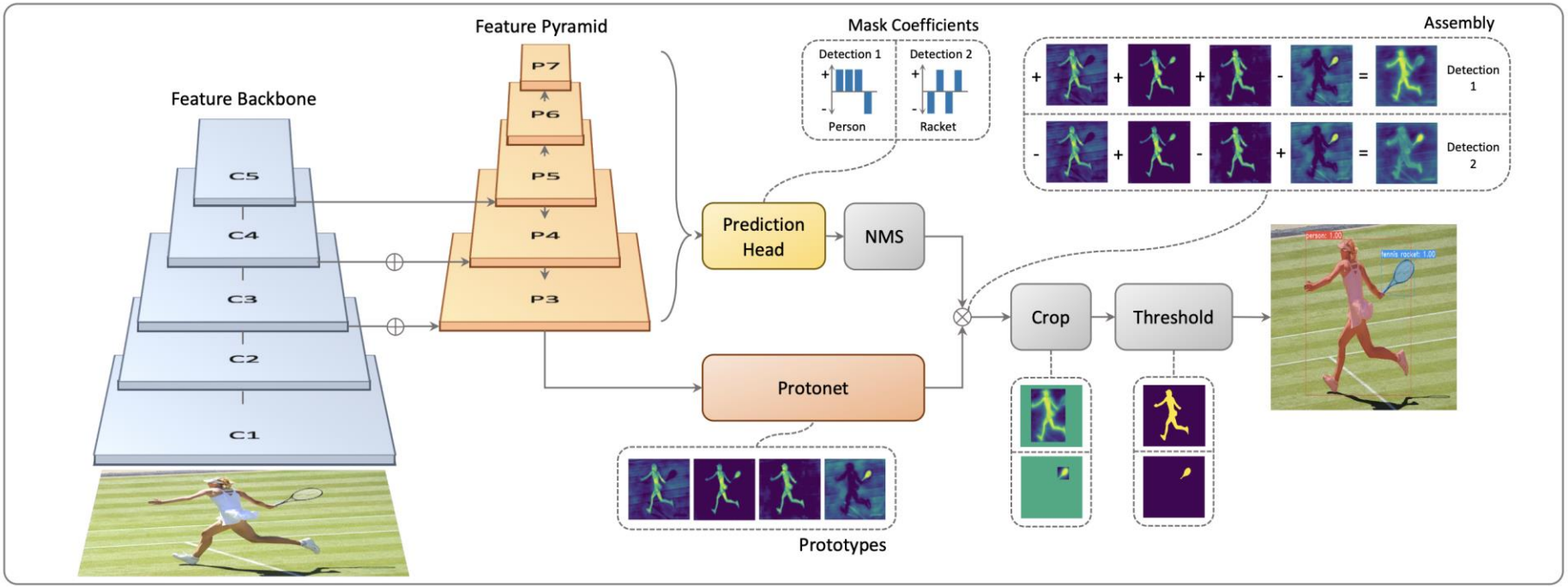
He et al, “Mask R-CNN”, ICCV 2017

Mask R-CNN Also does pose



He et al, “Mask R-CNN”, ICCV 2017

YOLACT – Faster option



Bolya et al, “YOLACT: Real-Time Instance Segmentation”, ICCV 2019